DS701 Homework 8: SVD and PCA

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Name: _____ **BU ID:** _____

How this works. Answer **by hand on paper** (or a tablet with a stylus), show your work, then scan and upload to Gradescope, matching each problem to its pages. This is deliberate practice for the closed-notes quizzes, which are in the same style — so do it as much as possible without a computer or an AI assistant; if stuck, try AI in tutorial style or ask course staff. **Participation-graded:** a serious attempt at every problem earns full credit; solutions are released the day after the deadline. **Budget under two hours:** seven short parts — a concept, a reading of printed output, or one line of arithmetic from a sheet formula; nothing decomposed by hand.

Covers: L12 (SVD and Low-Rank Approximation) and L13 (PCA and t-SNE). **Reference sheet:** quiz-2 sheet §12 and quiz-3 sheet §13; anything else is printed in the problem.

Problem 1: Reading a singular-value spectrum

A building has 300 temperature sensors, read hourly for 2,000 hours: $A \in \mathbb{R}^{2000\times 300}$ (rows = hours, columns = sensors). Its SVD $A = U\Sigma V^T$ has singular values $\sigma_1 = 40$, $\sigma_2 = 30$, $\sigma_3 = 10$, $\sigma_4 = 8$, $\sigma_5 = 6$, and $\sigma_6, \dots, \sigma_{300}$ all below 10^{-12}.

(a) What is the rank of A? One sentence on why the tail of tiny values does not count, and one on what this rank says about the 300 sensors (think of $A = CR$).

rank A = _____

(b) From the spectrum alone, fill in the blanks for $k = 1, 2, 3$. Then: what is the smallest k whose $A^{(k)}$ captures at least 95% of $\|A\|_F^2$, and for that k, how many numbers does the factored $A^{(k)}$ store, as a fraction of mn?

fraction of $\|A\|_F^2$ captured by $A^{(k)}$: k=1: _____ k=2: _____ k=3: _____

$\|A - A^{(k)}\|_F^2$: k=1: _____ k=2: _____ k=3: _____

k = _____ numbers stored for $A^{(k)}$: _____ fraction of mn: _____

Problem 2: What the singular vectors mean

Practices: reading the first two singular vectors of a small table as patterns / latent factors.

A café records how many of four products five regulars bought last month:

	coffee	filters	tea	honey
customer 1	8	6	1	0
customer 2	6	5	0	1
customer 3	1	0	7	6
customer 4	0	1	8	7
customer 5	4	3	4	3

Its SVD $A = U\Sigma V^T$ has $\sigma = (16.1,\ 12.3,\ 1.2,\ 0.8)$, and the first two singular-vector pairs are (rounded to two decimals; signs chosen so that $\mathbf{v}_1$ is positive and the first entry of $\mathbf{v}_2$ is positive):

$$\mathbf{v}_1 = \begin{bmatrix} 0.45 \\ 0.36 \\ 0.63 \\ 0.53 \end{bmatrix} \begin{matrix} \text{coffee} \\ \text{filters} \\ \text{tea} \\ \text{honey} \end{matrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 0.65 \\ 0.50 \\ -0.43 \\ -0.38 \end{bmatrix}, \qquad \mathbf{u}_1 = \begin{bmatrix} 0.40 \\ 0.31 \\ 0.50 \\ 0.56 \\ 0.43 \end{bmatrix} \begin{matrix} \text{c1} \\ \text{c2} \\ \text{c3}, \\ \text{c4} \\ \text{c5} \end{matrix} \qquad \mathbf{u}_2 = \begin{bmatrix} 0.63 \\ 0.49 \\ -0.38 \\ -0.46 \\ 0.10 \end{bmatrix}.$$

(a) In one sentence, what pattern in the products does $\mathbf{v}_1$ describe, and what does a customer's entry in $\mathbf{u}_1$ measure? Which customer has the largest first coordinate in $U'\Sigma'$ (the sheet's "row of $U'\Sigma'$")? Use the printed numbers — one multiplication at most.

Customer: _____

(b) Now the second pair. In one or two sentences, what *contrast* does $\mathbf{v}_2$ capture, and which customers does $\mathbf{u}_2$ put on each side of it? What does customer 5's entry 0.10 say about them? Finally: $\sigma_1^2 + \sigma_2^2$ is 99.5% of $\|A\|_F^2$ — in one sentence, what does that tell you about this table?

Problem 3: PCA variance accounting, preprocessing, and a t-SNE map

Practices: reading eigenvalues as variances, knowing why centering and scaling matter, and reading a t-SNE picture without over-reading it.

(a) A dataset has $m = 101$ rows and $n = 6$ features. The features are **standardized** (each column centered and scaled to variance 1) before PCA, and the covariance matrix S has eigenvalues

$$\lambda = (3.0,\ 1.5,\ 0.9,\ 0.3,\ 0.2,\ 0.1).$$

Fill in the table. What is the smallest number of components d that retains at least 90% of the variance? Why is the total T exactly 6 here, in one sentence? Lastly, if the centered, standardized data matrix X had been decomposed by SVD instead, what would σ_1 be?

PC i	λ_i	ratio λ_i/T	cumulative
1	3.0	_____	_____
2	1.5	_____	_____
3	0.9	_____	_____
4	0.3	_____	_____
5	0.2	_____	_____
6	0.1	_____	_____

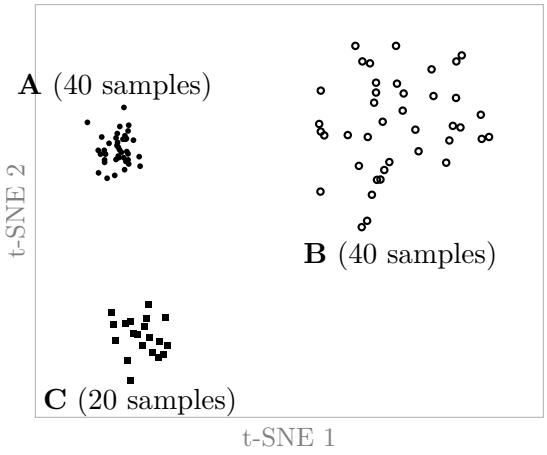
$T =$ _____ $d =$ _____ $\sigma_1 =$ _____

(b) True or false, with one reason each.

(i) A teammate's features are house price (dollars), age (years) and number of rooms. They say: "I centered every column, so PCA is fine — scaling is optional."

(ii) Another teammate skips centering and takes the top right singular vector of the raw X_0 as "PC1": "the top singular vector is the direction of maximum variance either way."

(c) The figure is a t-SNE map (perplexity 30, one random start) of 100 tissue samples, each originally described by 50 PCA features. Three groups are visible.



For each claim, say **supported** or **not supported** by this map, with one reason.

(i) "Group B's samples are more varied among themselves than group A's - look how much more spread out B is."

(ii) "Group A is biologically closer to group C than to group B, since A sits nearer to C."

(iii) "Two samples that sit next to each other inside group A are near neighbours in the original 50-dimensional space."

(iv) "Re-running t-SNE with a different random seed would reproduce this layout."

Finally: the team wants to train a classifier on a 20-number summary of each sample. Which method, PCA or t-SNE, should produce those 20 numbers, and why (one sentence)?